Literal descriptions in two semilinear projective tables

Private recovery component

Theorem 1. *In the exact current companion tables for* $\mathrm{P\Sigma L}_2(8)$ *and* $\mathrm{P\Gamma L}_2(27)$*, all 25 and 67 displayed subgroup structure descriptions hold for the actual representatives specified by the accepted numerical row bijections. In particular, the top groups have the displayed decompositions* $\mathrm{L}_2(8) : 3$ *and* $(\mathrm{L}_2(27) : 3) : 2$*. Each description asserts a decomposition, without asserting that factor names alone determine an isomorphism type.*

Source and exact notation. The tables are attributed to Thomas Connor and Dimitri Leemans. All four native pages were directly inspected. Their current bytes are retained; identity with unseen historical companion files is not asserted. The numerical subgroup inventories, class lengths, weighted incidences and row bijections are acquired inputs. The present result concerns the structural descriptions in these two tables only. No lattice-property premise is taken from the headings.

Section 3 of the acquired 2015 article specifies direct product $\times$, semidirect product :, nonsplit extension $\cdot$, and left association of products. Thus row 19 in the second table means $(2 \times 6) : S_3$, and row 13 means $(7 : 3 \times 2) \cdot 2$. The native ledger distinguishes 3^3, 3^2, 2^3 and 2^2 from cyclic orders. The standard notation 3^{1+2} is given its extraspecial meaning: a group of order 27 whose center, derived subgroup and Frattini subgroup coincide and have order 3. No exponent or extraspecial sign is specified by the source. Both exponent-three and exponent-nine groups occur, and the registry keeps their models distinct.

Actual groups and acquired identification. Every row is joined to literal embedded permutation generators in a saved complete inventory. The strict parser accepts only permutation cycles inside `Group`; no displayed subgroup name is executed as a constructor. The original top models were the recorded permutation images of $\mathrm{Aut}(\mathrm{PSL}(2, q))$, for $q = 8, 27$. Their exact ordered generators, executed drivers and named-model identification are retained from the accepted numerical component. The classical identification of these automorphism groups with the stated semilinear groups is an explicit inherited premise, not inferred from orders. The new certificates also verify the printed top decompositions.

Registry and transport. The accepted PSL_2 and PGL_2 components supply 36 actual models and already proved descriptions needed here. Multiple models of a description are retained whenever available. This packet adds 26 concrete models, with references only to acquired or earlier verified factors. The acquired marked generator-edge criterion MC.3 verifies each transport: the stored bijection sends identity to identity and satisfies $f(xa) = f(x)f(a)$ on every right-generator edge. Induction on words makes it a homomorphism, and bijectivity makes it an isomorphism. All maps are checked on literal arrays; no transient GAP isomorphism result is accepted without these checks. This uses order times generator count, rather than quadratic multiplication tables.

Lemma 1. *Let $H = \langle x, y\rangle$ have order 27, and put $z = x^{-1}y^{-1}xy$. Suppose z is central of order 3, $x^3, y^3 \in \langle z\rangle$, and $Z(H) = \langle z\rangle$. Then H is extraspecial.*

Proof. The quotient by $Z = \langle z\rangle$ is generated by two commuting order-dividing-three elements and has order nine, so it is C_3^2. Since z is a commutator and this quotient is abelian, $H' = Z$. Every maximal subgroup contains Z. Indeed, a subgroup of order nine not containing Z would have product with Z equal to H; that subgroup is abelian, and Z is central, contradicting $H' \neq 1$. A subgroup of order three not containing Z extends to its proper order-nine product with Z, so cannot be maximal. Consequently the maximal subgroups are exactly the inverse images of the four lines in H/Z. Their intersection is Z, proving $\Phi(H) = Z$. The elementary fact that groups of order p^2 are abelian follows from the class equation and the fact that a group with cyclic central quotient is abelian. □

The literal verifier checks every hypothesis, including the exact center by commuting with the two generators. Two new order-27 models have respective element-order spectra $\{1, 3\}$ and $\{1, 3, 9\}$. They account for the source's unqualified extraspecial descriptions without identifying two nonisomorphic groups.

Products and dotted extensions. For $H = N : K$, the acquired criterion checks literal membership, normality of N on generators, both factor descriptions, $N \cap K = 1$ and $|N||K| = |H|$. These imply the split decomposition; a noncommuting generator pair pays nontrivial action. For $H = N \times K$, the same product and intersection checks, together with commutation of every pair of factor generators, make $(n, k) \mapsto nk$ an isomorphism $N \times K \to H$. There are nine new direct-product models. Each centered-dot step has index two: the complete check $x^2 \neq 1$ for all outside elements proves that there is no complement. The sole new dotted model has 42 outside elements.

Finite certificate and controls. The witness-only GAP pass returned in 2.97 seconds, with all 92 row witnesses; it did not enumerate ambient subgroups. Normal-subgroup, complement and isomorphism routines only propose certificates. The literal verifier independently checks all 92 source joins, 26 new models and 137 transports, comprising 465,483 right-generator edges. Both top rows receive full certificates. Every proper subgroup generator is parsed in its actual ambient action; no smaller canonical action is substituted for an embedded subgroup.

The controls damage identity images, a bijective nonhomomorphism, element and row coverage, source row maps, actual generators, superscripts, the extraspecial atom, dot versus colon, product association, acquired models, quotient index, elementary basis orders, acyclic references, the commutator and a direct factor. Each must be rejected. Independent all-element product controls accept $C_3 \times C_2$ and reject the noncommuting factors in S_3. Both actual extraspecial spectra are checked, while the abelian groups C_{27} and $C_9 \times C_3$ are rejected. The inherited prime-index controls distinguish C_{p^2} from C_p^2 for $p = 2, 3$. Ordinary and optimized Python use the same explicit guards.

Coverage and attribution boundary. This is a proof of the displayed existence descriptions in 92 exact current rows, conditional on the acquired complete inventories, numerical bijections and named-model interfaces. It neither specifies a unique extension from its factors nor gives a new subgroup census. The extra registry models and their actual transports are the new finite evidence; standard naming conventions, prior atomic descriptions and the marked-edge lemma remain attributed inputs. Root suggested continued registry reuse; the component author prepared the native ledger, certificates, new literal tests and source joins. The whole atlas and unseen editions remain outside this component. No original-source, question, formalization or measurement count changes here.